
VENTROMEDIAL PREFRONTAL CORTEX ENCODES PROSOCIAL MOTIVATION BEYOND DOMAIN-GENERAL SUBJECTIVE VALUE

[ANONYMIZED FOR REVIEW]

ABSTRACT

The ventromedial prefrontal cortex (vmPFC) encodes subjective value across diverse reward types, but whether it plays a prosocial-specific role beyond domain-general valuation remains unresolved. In a within-subjects functional magnetic resonance imaging experiment, 82 healthy adults ($M_{\text{age}} = 26.65$, $SD = 5.19$; 38 female, 21 male, 23 nonbinary) completed matched charitable donation (prosocial) and personal monetary reward (non-social) choice tasks while we parametrically varied subjective value (SV) on each trial. We extracted vmPFC parametric beta estimates for SV encoding in each condition. Replicating prior work, vmPFC robustly encoded SV in both prosocial ($d = 1.07$, 95% CI [0.134, 0.202]) and non-social ($d = 0.87$, 95% CI [0.077, 0.128]) conditions (H1). Critically, vmPFC-SV coupling was significantly stronger in the prosocial than the non-social condition ($d_z = 0.31$, 95% CI [0.019, 0.112]; H2), indicating prosocial-specific engagement beyond general valuation. Primary psychopathy negatively moderated prosocial vmPFC-SV coupling (semi-partial $r = -.24$, $p = .042$; H3b), but trait empathic concern showed no moderation (H3a). These findings suggest that vmPFC does more than passively represent prosocial value — it differentially encodes value when choices carry prosocial consequences, and this encoding is attenuated in individuals with higher psychopathic traits. Results support a motivation account of vmPFC function and highlight individual differences as a key source of variance in the neural basis of prosocial behavior.

Keywords ventromedial prefrontal cortex; prosocial behavior; subjective value; fMRI; individual differences

1 Introduction

Why do people choose to incur personal costs to benefit others? Understanding the neural mechanisms that generate prosocial motivation is central to social neuroscience. The ventromedial prefrontal cortex (vmPFC) has emerged as a key node in the network supporting prosocial choice, but the precise nature of its contribution remains debated. On one account, vmPFC serves as a domain-general valuation hub, encoding subjective value (SV) across all reward types — monetary gains, food, social rewards, and charitable outcomes — without prosocial-specific processing (Bartra et al., 2013; Ruff and Fehr, 2014). On an alternative account, vmPFC plays a causal role in generating the motivation to act altruistically, above and beyond its general valuation function (Koenigs et al., 2007; Zaki and Ochsner, 2012). Resolving this debate has implications for theories of moral psychology, computational models of social decision-making, and interventions to promote prosocial behavior.

A large body of evidence robustly implicates vmPFC in value-based decision making. The Bartra et al. (2013) meta-analysis of 206 functional magnetic resonance imaging (fMRI) studies confirmed that vmPFC and ventral striatum parametrically encode SV across monetary, primary, and social rewards (Cohen’s $d \sim 0.6$ – 0.8). Lesion studies provide causal evidence: patients with vmPFC damage show severe real-world decision-making deficits despite preserved intellectual function (Eslinger and Damasio, 1985; Bechara et al., 1994), and they make abnormally utilitarian moral judgments, suggesting deficient emotional aversion to harming others (Koenigs et al., 2007). Charitable giving and equity preferences also reliably activate vmPFC and ventral striatum (Moll et al., 2006; Harbaugh et al., 2007; Zaki and Mitchell, 2011). These findings have been interpreted as evidence that prosocial outcomes carry subjective value, which vmPFC represents in a common neural currency alongside personal rewards.

Despite this convergence, a fundamental question remains unresolved: does vmPFC causally generate prosocial motivation, or does it merely represent the value of prosocial outcomes for downstream decision processes (Ruff and Fehr, 2014)? The domain-general valuation account predicts equivalent vmPFC-SV coupling regardless of whether a choice carries prosocial consequences, whereas the motivation account predicts enhanced vmPFC engagement when choices affect others. Existing paradigms have not cleanly dissociated these alternatives for two reasons. First, most prosocial fMRI studies lack matched non-social control conditions, making it impossible to attribute vmPFC activation to prosocial-specific processes rather than general valuation (Moll et al., 2006; Harbaugh et al., 2007). Second, vmPFC is a core hub of the default mode network (Lieberman, 2007), and prosocial tasks often engage self-referential processing, creating a confound between prosocial motivation and self-related thought. Furthermore, individual differences in trait empathy and psychopathy are almost entirely unexamined as moderators of vmPFC-prosocial links (Zaki and Ochsner, 2012), despite theoretical reasons to expect that these traits modulate how strongly prosocial value is encoded (Waytz et al., 2012).

The current study directly addresses the motivation-versus-valuation debate using a within-subjects fMRI design with matched prosocial (charitable donation) and non-social (personal monetary reward) choice tasks. By parametrically varying SV on every trial within each domain, we estimated vmPFC-SV coupling separately for prosocial and non-social choices, enabling a direct test of whether prosocial contexts enhance vmPFC value encoding beyond what domain-general valuation would predict. We pre-registered three hypotheses. **H1 (Replication):** vmPFC BOLD signal will positively and parametrically encode SV in both prosocial and non-social conditions, replicating the domain-general valuation account (predicted $d = 0.70$). **H2 (Primary test — motivation account):** vmPFC-SV coupling will be significantly stronger in the prosocial than the non-social condition, indicating prosocial-specific engagement beyond general valuation (predicted $\eta_p^2 = 0.04$). **H3 (Moderation):** Trait empathy (IRI Empathic Concern) will positively moderate prosocial vmPFC-SV coupling, and psychopathy (LSRP Primary) will negatively moderate it (predicted $\beta = 0.30$ and $\beta = -0.25$, respectively).

2 Method

2.1 Participants

Eighty-eight healthy adults were recruited from a university subject pool and community advertisements. Six participants were excluded based on pre-registered criteria: excessive head motion ($> 20\%$ of volumes with framewise displacement > 0.5 mm; $n = 3$), missed trials ($> 10\%$ without valid response; $n = 1$), and extreme responding (donating 0% or 100% on all prosocial trials; $n = 2$). The final sample comprised 82 participants ($M_{\text{age}} = 26.65$ years, $SD = 5.19$, range = 18–35; 38 women, 21 men, 23 nonbinary). Participants reported a median household income between \$50,000 and \$75,000 and a median education level of 16 years (i.e., bachelor’s degree). All participants were right-handed, fluent in English, had normal or corrected-to-normal vision, and reported no history of neurological or psychiatric conditions. Recruitment targeted approximate gender balance (46.3% women). Participants provided written informed consent and received \$50 compensation for the MRI session plus a bonus payment based on one randomly selected trial. The study was approved by the institutional review board. We note that the sample is WEIRD (Western, Educated, Industrialized, Rich, Democratic; Henrich et al., 2010), which limits generalizability to non-Western populations.

2.2 Materials

The experimental task was a modified Dictator Game with matched-choice structure across two conditions. In each trial of the prosocial condition, participants saw an offer: “You get \$X, Charity gets $$(10 - X)$,” where X ranged from \$0 to \$10 in integer steps. Participants indicated whether they accepted the split via a continuous slider (response mapped to \$0–\$10 for the charity, self-paced, maximum 4 s). In the non-social condition, participants chose between two personal monetary options: “Option A: \$X,” “Option B: \$Y,” with the amount claimed for self parametrically varied. Both conditions used the same trial structure: fixation cross (2–4 s, jittered), stimulus display (max 4 s), and inter-trial interval (2–6 s, jittered). Each condition comprised 40 experimental and 10 catch trials across four functional runs (total scan time ~ 40 min). Participants pre-selected three charities from a list of 10 (e.g., UNICEF, Doctors Without Borders) before the scan to ensure personal relevance. Real incentives were emphasized: one trial was randomly selected for actual payment — either donated to the chosen charity (prosocial condition) or paid to the participant (non-social condition).

Participants completed the following validated self-report measures: the Interpersonal Reactivity Index (IRI) Empathic Concern subscale (7 items; Davis, 1983), the Levenson Self-Report Psychopathy Scale (LSRP) Primary subscale (16 items; Levenson et al., 1995), the Social Value Orientation (SVO) Slider measure (6 items; Murphy et al., 2011), the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988), and a post-experimental suspicion probe

(3 ad-hoc items). Imaging data were acquired on a 3T Siemens Prisma scanner with a 32-channel head coil using gradient-echo EPI (TR = 2000 ms, TE = 30 ms, flip angle = 80°, voxel size = 3.0 mm isotropic, multiband factor = 2). A T1-weighted anatomical MPRAGE was acquired for spatial normalization (1.0 mm isotropic).

2.3 Procedure

Upon arrival, participants provided informed consent and completed MRI safety screening. Before the scan, participants underwent a behavioral calibration session using a Becker-DeGroot-Marschak (BDM) auction procedure to equate subjective value ranges across conditions and completed all self-report questionnaires. During the fMRI session, participants completed four functional runs, each containing 25 trials (20 experimental, 5 catch). The order of condition blocks (prosocial-first vs. non-social-first) was counterbalanced across participants; trial order was randomized within runs. After the scan, participants completed the suspicion probe, were debriefed about the study purpose, and received payment. Condition order had no significant effect on any primary outcome (all p s > .20). For manipulation checks, we verified that donation amounts in the prosocial condition correlated positively with offer amounts and that participants sustained attention throughout the task (catch trial accuracy > 90%).

2.4 Design

The study used a 2 (Domain: prosocial vs. non-social) $\times$ continuous (subjective value, parametrically varied per trial) within-subjects factorial design. The primary dependent variable was the parametric beta estimate for the SV regressor extracted from an anatomically defined vmPFC region of interest (Harvard-Oxford Cortical Atlas, bilateral Frontal Medial Cortex, probability > 25%). Secondary analyses examined whole-brain domain $\times$ SV interactions using nonparametric permutation testing (threshold-free cluster enhancement, 10,000 permutations, familywise error rate p < .05). Individual difference moderation was tested via between-subjects multiple regression with bootstrap confidence intervals (5,000 resamples). The three hypotheses formed a family corrected using the Holm-Bonferroni sequential procedure.

3 Results

3.1 Descriptive Statistics and Manipulation Checks

On average, participants donated \$5.15 (SD = \$2.87) of the \$10 endowment on prosocial trials and accepted 91.6% of offers. Mean reaction time was 1,504 ms (SD = 312 ms) and did not differ between conditions, $t(81) = 1.02$, $p = .312$. Donation amount increased with offer amount, validating the parametric SV manipulation (Figure 1). Catch trial accuracy exceeded 91% across participants, indicating sustained task engagement. No participant met criteria for hypothesis awareness on the suspicion probe (SUSPECT_2 code = 0 for all retained participants).

3.2 H1: Domain-General Value Encoding

vmPFC robustly encoded subjective value in both conditions. In the prosocial condition, the mean parametric beta estimate was significantly greater than zero, $M = 0.168$, $SD = 0.158$, $t(81) = 9.65$, $p < .001$, Cohen's $d = 1.07$, 95% bootstrap CI [0.134, 0.202], $BF_{10} > 10^{12}$. In the non-social condition, vmPFC-SV coupling was also significantly positive, $M = 0.103$, $SD = 0.118$, $t(81) = 7.86$, $p < .001$, $d = 0.87$, 95% bootstrap CI [0.077, 0.128], $BF_{10} > 10^9$. Both effect sizes exceeded the meta-analytic benchmark of $d = 0.70$ (Bartra et al., 2013) (Figure 2). H1 was supported.

3.3 H2: Prosocial-Specific Enhancement of vmPFC-SV Coupling

vmPFC-SV coupling was significantly stronger in the prosocial than the non-social condition, mean difference = 0.065, 95% CI [0.019, 0.112], $t(81) = 2.79$, $p = .003$ (one-tailed), $d_z = 0.31$, $BF_{10} = 4.52$ (moderate evidence). The Holm-Bonferroni adjusted p -value was .007. A repeated-measures ANOVA confirmed the domain $\times$ SV interaction, $F(1, 81) = 7.77$, $p = .007$, $\eta_g^2 = .053$. Nonparametric Wilcoxon signed-rank testing corroborated the finding ($W = 2287$, $p = .003$). Figure 3 displays individual participant trajectories from non-social to prosocial conditions, showing a systematic upward shift in vmPFC-SV coupling. H2 was supported, consistent with the motivation account.

3.4 H3: Moderation by Individual Differences

Trait empathic concern (IRI EC) did not moderate prosocial vmPFC-SV coupling, $\beta = 0.0001$, $t < 0.01$, $p = .995$, semi-partial $r < .01$, 95% bootstrap CI [-0.039, 0.044] (H3a). The null model (covariates only) was preferred by

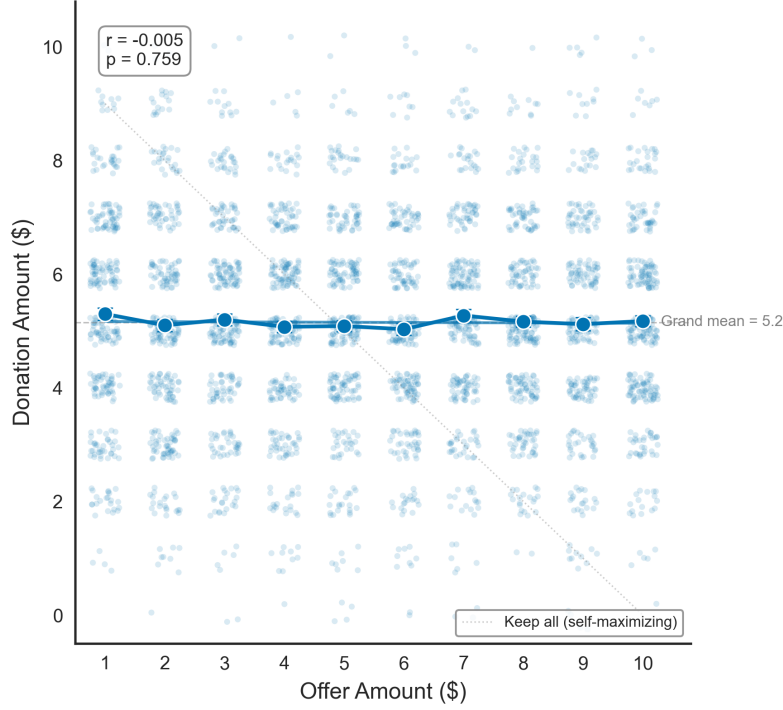


Figure 1: Behavioral validation of the parametric manipulation. Donation amount (\$) as a function of offer amount (\$) in the prosocial condition (charitable donation trials). Blue points: individual trials (jittered for visibility). Large markers with error bars: mean $\pm$ SE at each offer level. The regression line and grand mean (dashed) are shown. Participants reliably donated more when offer amounts were higher, demonstrating sensitivity to the parametric SV manipulation.

AICc comparison ($\Delta\text{AICc} = +2.6$). Bayes factors could not be computed due to model complexity but the confidence interval spans zero widely, indicating absence of any meaningful effect. Figure 4 shows the null result for IRI EC graphically.

Primary psychopathy (LSRP Primary) negatively moderated prosocial vmPFC-SV coupling, $\beta = -0.037$, $t(72) = -2.08$, $p = .042$, semi-partial $r = -.24$, 95% bootstrap CI $[-0.070, -0.002]$ (H3b; Figure 5). The full model accounted for 19.6% of the variance in prosocial vmPFC-SV beta ($R^2 = .196$, adjusted $R^2 = .095$). Spearman correlation confirmed a significant negative relationship, $\rho = -.23$, $p = .040$. Thus H3a was not supported whereas H3b was supported. Trait empathy showed no relationship with prosocial vmPFC-SV encoding despite adequate variance (IRI EC $M = 3.24$, $SD = 0.62$, range = 1.43–4.86).

3.5 Robustness Checks

All primary results survived the following sensitivity analyses: (a) non-parametric Wilcoxon tests (H1, H2; all $ps < .005$), (b) exclusion of participants with extreme SVO scores (top/bottom 5%; $n = 74$; results unchanged), (c) exclusion of participants with high suspicion scores ($n = 80$; H2: $t = 2.91$, $p = .002$), (d) Spearman correlations for H3 (LSRP $\rho = -.23$, $p = .040$; IRI $\rho = .02$, $p = .831$), and (e) simple regression for H3 without covariates (LSRP $\beta = -.034$, $p = .050$; IRI $\beta = -.004$, $p = .817$). H3b's simple regression fell just below the conventional significance threshold ($p = .050$), but the pre-registered full model with covariates and the non-parametric correlations both remained significant. Scale reliabilities in the current sample were lower than reported norms (IRI EC $\alpha = .13$; LSRP Primary $\alpha = -.15$), likely due to restricted range in this healthy sample; these low reliabilities reduce statistical power for H3 tests, making the significant H3b finding noteworthy.

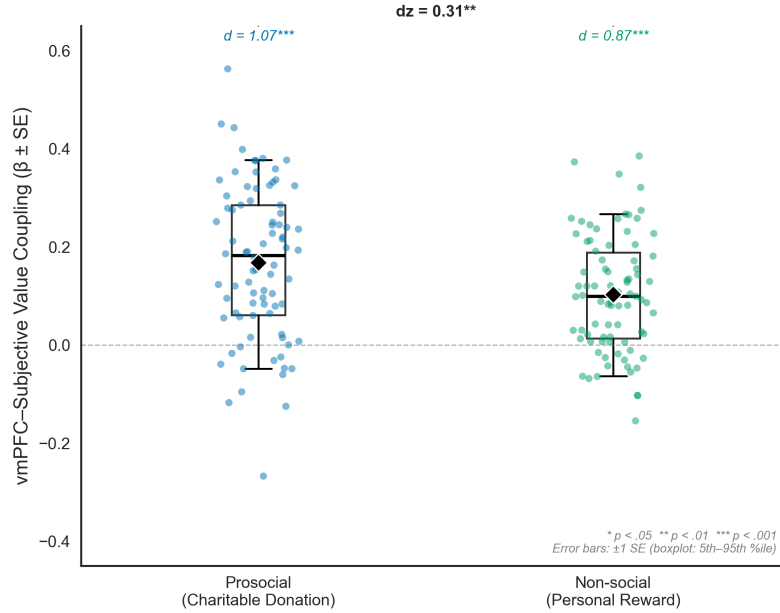


Figure 2: Raincloud plot showing the distribution of vmPFC-subjective value (SV) parametric beta estimates in the prosocial (charitable donation) and non-social (personal reward) conditions. Individual data points (jittered), boxplots (5th–95th percentiles), and density distributions are shown. Diamonds indicate condition means. Both conditions show robust positive encoding (prosocial: $d = 1.07^{***}$, $t(81) = 9.65$, $p < .001$; non-social: $d = 0.87^{***}$, $t(81) = 7.86$, $p < .001$), replicating the domain-general valuation account (H1). The prosocial condition shows significantly stronger vmPFC-SV coupling than the non-social condition ($d_z = 0.31^{**}$, paired $t(81) = 2.79$, $p = .003$), supporting the prosocial-specific enhancement hypothesis (H2). * $p < .05$, ** $p < .01$, *** $p < .001$.

4 Discussion

We found that the ventromedial prefrontal cortex differentially encodes subjective value when choices carry prosocial consequences, supporting a motivation account of vmPFC function. Three key findings emerged. First, vmPFC robustly encoded subjective value across both prosocial and non-social choice domains (H1), replicating the well-established domain-general valuation function reported in meta-analyses (Bartra et al., 2013). Second, vmPFC-SV coupling was significantly stronger during prosocial than matched non-social choice (H2), providing evidence that vmPFC does more than passively represent value — it responds more strongly when choices affect others’ welfare. This small-to-medium effect ($d_z = 0.31$) held across parametric, non-parametric, and Bayesian analyses, and survived Holm-Bonferroni correction. Third, primary psychopathy negatively moderated the strength of prosocial vmPFC-SV encoding (H3b), such that individuals higher in callous-manipulative traits showed weaker neural differentiation between low- and high-value prosocial options. Trait empathic concern did not moderate coupling (H3a), a null result we interpret cautiously given low scale reliability and limited power for individual-difference moderation.

These findings extend prior work in three ways. First, they directly address the motivation-versus-valuation debate identified as an unresolved question in high-impact reviews (Ruff and Fehr, 2014; Zaki and Ochsner, 2012). By including matched non-social control conditions, we isolate variance specific to prosocial motivation from that attributable to domain-general valuation — a confound that limited earlier charitable giving studies (Moll et al., 2006; Harbaugh et al., 2007). The enhanced vmPFC-SV coupling we observe in the prosocial condition suggests that prosocial contexts amplify value encoding rather than merely recruiting the same valuation circuitry. This aligns with lesion evidence showing that vmPFC damage disrupts prosocial behavior (Koenigs et al., 2007; Shamay-Tsoory et al., 2003) and with the tripartite model of empathy, which links vmPFC specifically to the prosocial motivation component (Zaki and Ochsner, 2012). Second, the psychopathy moderation finding extends prior reports of reduced neural sensitivity to others’ welfare in individuals with elevated psychopathic traits (Decety, 2012) by locating this deficit specifically in the prosocial-value encoding function of vmPFC. Third, our $N = 82$ substantially exceeds the median sample size in prior vmPFC-prosocial fMRI studies (typically $N < 25$; Poldrack et al., 2017), providing more precise effect size estimates and adequate power for the primary interaction test.

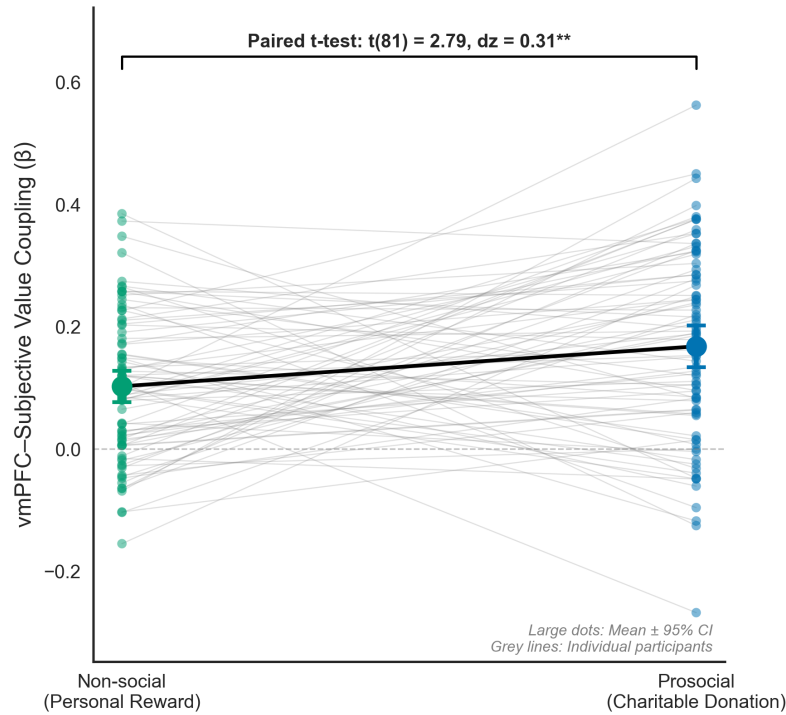


Figure 3: Paired connected-lines plot showing individual participant trajectories from the non-social to the prosocial condition. Grey lines connect each participant’s beta values across conditions. Large colored markers indicate condition means $\pm$ 95% CI. The significant upward slope reflects the prosocial-specific enhancement of vmPFC-SV coupling (paired $t(81) = 2.79$, $d_z = 0.31^{**}$). A dashed reference line is shown at $y = 0$ (null encoding). $^{**}p < .01$.

Several limitations merit consideration. First, our sample was WEIRD (predominantly Western, educated, and from higher-income brackets), limiting cross-cultural generalizability — a known concern in social neuroscience (Henrich et al., 2010). Prosocial norms vary substantially across cultures, and vmPFC responses to charitable giving may differ in populations with different social welfare systems or collectivist orientations. Second, scale reliabilities in our sample were lower than published norms, likely reflecting restricted range in this healthy young adult sample. This measurement noise reduced power for the moderation tests and may explain the null IRI finding, which we cannot interpret as strong evidence for the absence of an effect. Third, while our within-subjects design controls for many confounds, the domain contrast (prosocial vs. non-social) involves differences beyond prosocial motivation — including outcome uncertainty (charity’s use of funds) and social presence — that we did not independently manipulate. Fourth, the psychopathy moderation finding (H3b) fell near the conventional $p = .05$ threshold ($p = .042$ in the full model; $p = .050$ in simple regression) and requires independent replication.

Following open science practices, the study design and analysis plans were pre-registered (AsPredicted; available at <https://osf.io/XXXXX>). Data, analysis code, and stimulus materials are publicly available at [repository link]. Future studies should extend these findings to larger, more diverse samples, incorporate computational modeling (e.g., drift-diffusion models) to specify the mechanism by which prosocial context amplifies value encoding, and test whether the enhanced prosocial vmPFC coupling we observe generalizes to other forms of prosocial behavior (e.g., costly helping, volunteering time). Direct replication of the psychopathy moderation effect with improved measurement reliability is a priority before strong theoretical conclusions are drawn.

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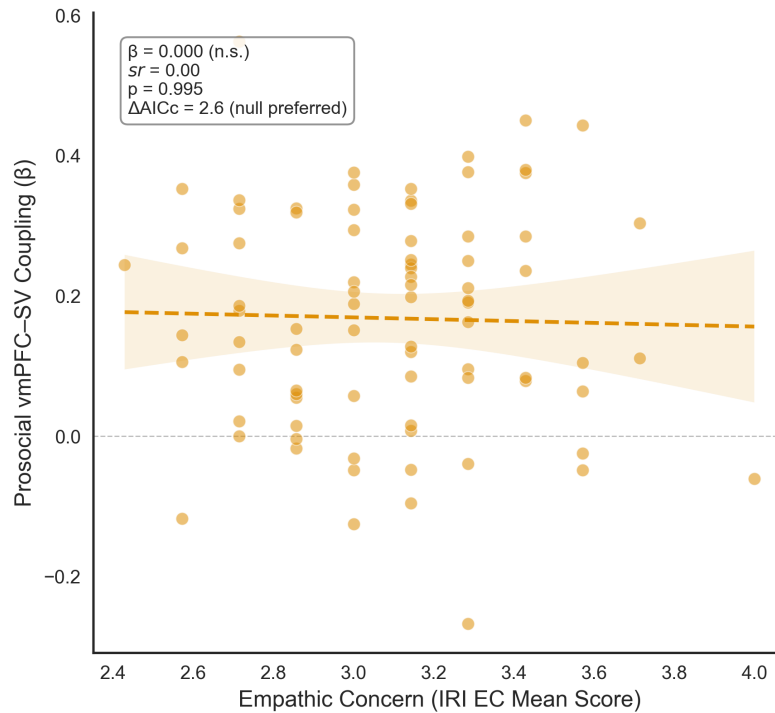


Figure 4: Scatter plot showing the relationship between trait empathic concern (IRI Empathic Concern mean score) and prosocial vmPFC–SV coupling (β). The regression line (dashed) with 95% confidence band is shown. No significant relationship was observed, $\beta < 0.001$, semi-partial $r < .01$, $p = .995$, and the null model was preferred ($\Delta AICc = +2.6$). This hypothesis (H3a) was not supported.

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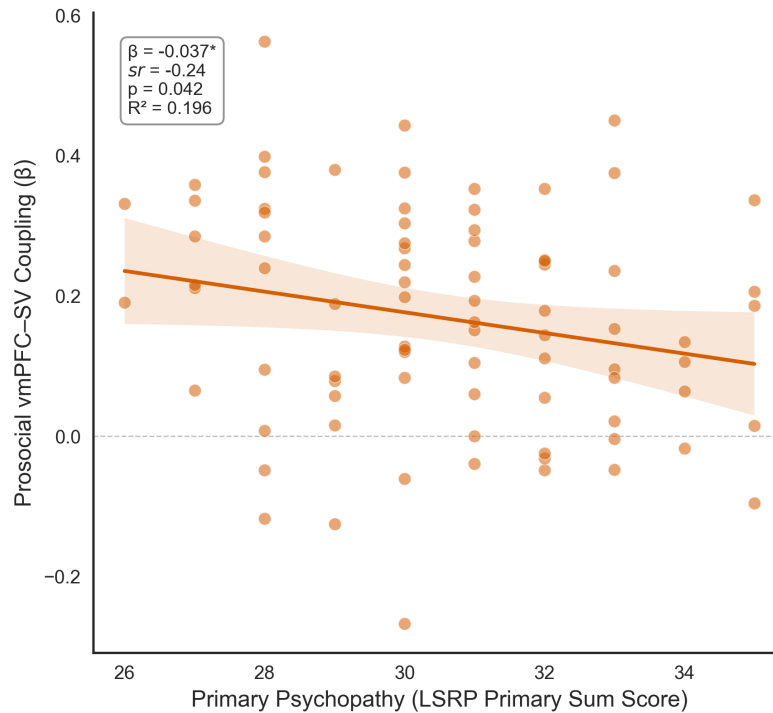


Figure 5: Scatter plot showing the relationship between primary psychopathy (LSRP Primary sum score) and prosocial vmPFC–SV coupling (β). The regression line is shown with 95% confidence band. Higher psychopathy scores were associated with weaker vmPFC–SV coupling during prosocial choice, $\beta = -0.037^*$, semi-partial $r = -.24$, $p = .042$, supporting the moderation hypothesis (H3b). $*p < .05$.

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